



University
Mohammed VI
Polytechnic



Deliverable #: lab2 (e.g.: Logical Design)

Data Management Course
UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	
Member 1	TAHA Yassine
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Member 4	Achraf RHANNAM
Repository Link	https://github.com/MrYassine18/data-management-labs.git

1 Team Members' Roles

M1: Person/Employee hierarchy, ISA relationships, Report-To, Laboratories.

M2: Projects, Budgets, Reservations, and approvals.

M3: Equipment, Certifications, Maintenance, and Calibration.

M4: Consumables and related relationships.

2 Key Challenges

2.1 M1 :

The main challenge was the multiple-level ISA hierarchy, especially Employee → Academic → Faculty and Employee → Non-Academic → Administrative/Technical. We solved this by creating a table for each subclass, using the inherited identifier as both primary key and foreign key.

Another challenge was the "at least one" participation constraint between Academic employees and Students. Since it could not be directly enforced using the basic course conversion rules, we represented the relationship with a junction table, Advises. The minimum participation itself would require an additional constraint such as a trigger or application-level check.

2.2 M2:

We represented Manages and MadeBy/For using foreign keys instead of separate tables, since each budget line has one manager and each reservation has one submitter and one research project.

2.3 M3:

These tables reference Person, Laboratory and Technical, so all team members used the entity and attribute names from the Lab 1 correction to keep the keys consistent. Some requirements, such as EQ7 and CE7, depend on reservations and time, so they cannot be enforced using keys only.

2.4 M4:

Figuring out the Monitors aggregation. Since the rules say exactly one tech staff monitors a stock (CO8), I avoided making an extra table. I just put tech-pid (as a foreign key) and monitoring into the Stocks table

3 Conversion Choices

3.1 M1 :

ISA relationships were converted using separate tables for each subclass. Many-to-many relationships were represented using junction tables, such as Advises and Attached. The recursive Report-to relationship was represented using a self-referencing foreign key

3.2 M2:

ResearchProject is a strong entity with its own table. Many-to-many relationships such as Participates, Reserves, FundsPrj, and FundsLab were implemented using junction tables with composite primary keys. Approves was kept as a separate table because approval is optional.

3.3 M3:

InstanceOf and LocatedIn were merged into EquipmentUnit as two foreign keys that cannot be null. Holds and Requires were converted into separate tables. The key of Holds is the person and the certification, so only the latest award is kept. Maintenance and CalibrationRecord are weak entities, identified by the serial number of the unit and their own partial key, and they are deleted with the unit. DoneBy was merged into Maintenance as the column doneBy, which stores the pid of the technical staff member.

3.4 M4:

Consumable and Supplier are strong entities, so they get their own simple tables. Supplies is a ternary relationship, so its primary key is a composite of all three foreign keys. Consumes is a many-to-many relationship, so it became a junction table with a composite key of (resId, consId).

4 Assumptions

4.1 M1 :

- Each person, employee, and student has a unique ID.
- An employee can have at most one supervisor.
- A laboratory can have at most one faculty supervisor.
- An equipment unit belongs to at most one laboratory.
- Faculty, Administrative, and Technical have a position attribute.

4.2 M2:

We assumed that each budget line has one academic manager, each reservation is made by one person for one research project, and many-to-many relationships use junction tables. Approval is optional, while additional business constraints are handled separately.

4.3 M3:

The maintenance start is stored as a date, so a unit has at most one maintenance per day. The validity period is in months. The remaining rules are written as extra constraints at the end of the SQL file.

4.4 M4:

We assumed that Consumables and Suppliers are independent entities, each Stock is monitored by exactly one technical staff member, and Supplies and Consumes represent their respective relationships. Some constraints, such as total participation, cannot be enforced using only primary and foreign keys.